

COMP3240 REINFORCEMENT LEARNING REPORT

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<https://github.com/Ziqingxue0927/220241971-RL-Project-Yunfei-Xue>

1. INTRODUCTION

In this project, we investigate Q-learning, for finding an optimal policy in a discrete Markov Decision Process (MDP). We apply Q-learning to a Grid-World navigation task, where an agent must learn to reach high-reward goal locations while avoiding terminal cliff states. To achieve this, we adopt the standard Q-learning update rule. The algorithm iteratively updates state-action values based on observed rewards and transitions, without requiring knowledge of the environment's transition probabilities.

The **SARMA** (State–Action–Reward–State–Action) updated rule, is an example of a Temporal Difference (TD) Learning algorithm, derived from the Bellman equation in the lecture 6. The formula as follow is

$$(1) \quad \Delta Q(s, a) = \eta (r + \gamma Q(s', a') - Q(s, a))$$

where

- $Q(s, a)$: The current estimate of the action value function for state s and action a .
- r : The reward received after taking action a in state s .
- s' : The next state resulting from taking action a in state s .
- a' : The next action selected in state s' , used in some on-policy algorithms.
- $Q(s', a')$: The estimated value of the next state-action pair.
- $\gamma \in [0, 1)$: The discount factor, which controls how much future rewards are taken into account.
- η : The learning rate, which controls how much newly acquired information overrides the old value.
- $\Delta Q(s, a)$: The update to be applied to $Q(s, a)$.

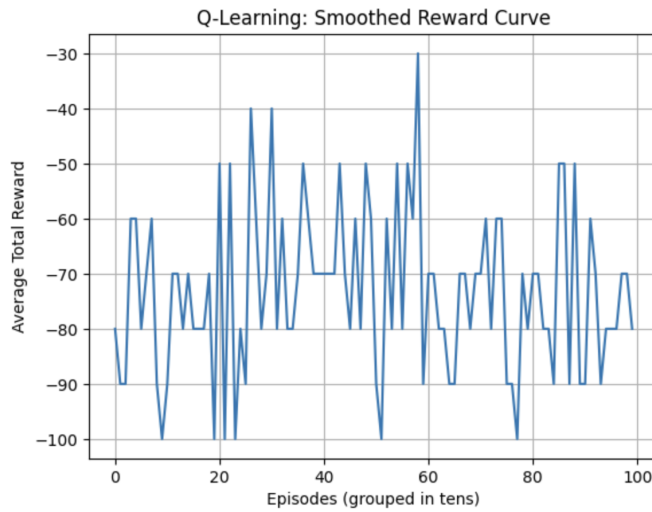
2. LEARNING CURVES

The code for Section 4.2 was heavily revised with the help of ChatGPT, particularly to implement repeated training runs and averaged performance plots, which were not achieved in the initial version.

The two plots show the agent's learning progress over 1000 episodes of Q-learning, with each curve representing a 10-episode moving average to reduce fluctuations. Although the output resembles the smoothing process

used in an $MA(q)$ model for non-stationary time series, it is not a true moving average process in the statistical sense.

The first plot shows the smoothed total reward averaged per 10 episodes under an ϵ -greedy exploration policy. Most of the rewards fluctuate between -70 and -100, indicating frequent penalties, likely due to the agent falling into cliff states. Occasional spikes suggest that the agent occasionally finds better paths, but overall, the learning has not fully converged.



The second plot shows the smoothed number of steps averaged over every 10 episodes. Similar to the first plot, it appears to converge toward an expected value, but with high variance. These fluctuations imply that convergence is slow, possibly due to sparse rewards or the presence of multiple goals and cliff states in the environment.



3. THE IMPACT OF HYPERPARAMETERS ON LEARNING PERFORMANCE

Richard S. Sutton discusses the impact of hyperparameters in his book *Reinforcement Learning: An Introduction*. In Chapter 2, *Multi-armed Bandits* (pp. 30–31), he compares ϵ -greedy methods with ϵ set to 0.01, 0.1, and 0, respectively. He remarks that “the ϵ -greedy methods eventually performed better because they continued to explore and to improve their chances of recognising the optimal action.” A bigger ϵ means the agent explores more at the beginning, which can make learning unstable. A smaller ϵ makes the agent learn more slowly at first, but it usually ends up with a better and more stable policy.

In Chapter 3, *Returns and Episodes* (p. 55), Sutton discusses the impact of the discount factor:

If $\gamma < 1$, the infinite sum in (3.8) has a finite value as long as the reward sequence $\{R_k\}$ is bounded. If $\gamma = 0$, the agent is “myopic” in being concerned only with maximizing immediate rewards. . . . But in general, acting to maximize immediate reward can reduce access to future rewards so that the return is reduced. As γ approaches 1, the return objective takes future rewards into account more strongly; the agent becomes more farsighted.

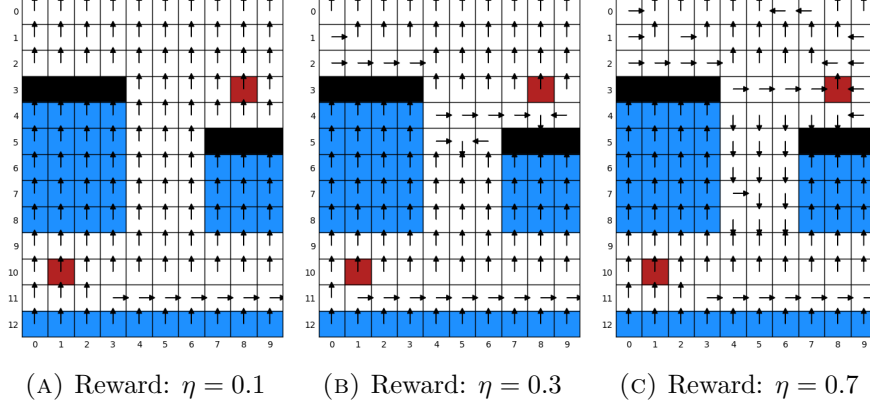
Sutton uses α to represent the learning rate in his book, which differs from the notation used in the lecture notes. In Chapter 6, *Temporal-Difference Learning* (p. 134), Sutton states:

“Expected Sarsa can safely set $\alpha = 1$ without suffering any degradation of asymptotic performance, whereas Sarsa can only perform well in the long run at a small value of α , at which short-term performance is poor.”

4. POLICY VISUALISATION

Following Sutton’s guidance in *Reinforcement Learning: An Introduction*, I adjusted the values of three key hyperparameters to improve training performance. Specifically, I set the discount factor $\gamma = 0.99$ and the learning rate $\eta = 0.1$ to promote long-term reward maximisation and ensure stable updates to the Q -values. The number of episodes was fixed at 1000.

In addition, I experimented with different values of the learning rate ($\eta = 0.1, 0.3$, and 0.7) to examine its influence. Since the environment is relatively small but complex—with multiple cliffs, walls, and sparse rewards—the learning rate plays a crucial role in determining how quickly and effectively the agent can adapt. A higher learning rate encourages faster learning and broader exploration during the early stages, helping the agent avoid getting stuck in suboptimal strategies and increasing the likelihood of discovering optimal paths to high-reward states.



Comparison of total rewards under different learning rates.

From the output, it is clear to see as learning rate η increases, the agent appears to learn a more aggressive path toward high-reward states. However, due to the constrained environment with cliffs and obstacles, a larger η may also increase the probability of falling into cliffs, revealing a trade-off between fast convergence and risk of instability.

5. CHALLENGES OF MULTIPLE GOAL STATES

In task 4, we visualise the different rewards under the different learning rates. The addition of a second reward location introduces a significant challenge in the agent's learning process. While the smaller reward at (2, 1) is closer and easier to find during early exploration, the higher reward of 5000 at (9, 8) is more distant and situated beyond dangerous cliff areas and walls.

It is clear to see that when η is low, the system has no tendency biased towards any rewards, but when increasing the η , as training progresses, more arrows point towards (9, 8) despite the high risk. This phenomenon elucidates the agent's ability to balance exploitation of short-term rewards with long-term value maximisation under the learning rate adjustment.